

MODULE 4 – DBMS (Lab Answers)

```
CREATE DATABASE school_db;
USE school_db;
```

```
CREATE TABLE students (
    student_id INT PRIMARY KEY,
    student_name VARCHAR(50),
    age INT,
    class VARCHAR(20),
    address VARCHAR(100)
);
```

```
INSERT INTO students VALUES
(1, 'Amit', 12, '7A', 'Ahmedabad'),
(2, 'Riya', 11, '6B', 'Surat'),
(3, 'Karan', 13, '8A', 'Rajkot'),
(4, 'Neha', 12, '7B', 'Vadodara'),
(5, 'Vivek', 14, '9A', 'Ahmedabad');
```

```
SELECT * FROM students;
```

```
SELECT student_name, age FROM students;
SELECT * FROM students WHERE age > 10;
```

```
CREATE TABLE teachers (
    teacher_id INT PRIMARY KEY,
    teacher_name VARCHAR(50) NOT NULL,
    subject VARCHAR(50),
    email VARCHAR(100) UNIQUE
);
```

```
ALTER TABLE students ADD teacher_id INT;
```

```
ALTER TABLE students
ADD FOREIGN KEY (teacher_id) REFERENCES teachers(teacher_id);
```

```
CREATE TABLE courses (
    course_id INT PRIMARY KEY,
    course_name VARCHAR(50),
    course_credits INT
);
```

```
ALTER TABLE courses ADD course_duration INT;
ALTER TABLE courses DROP COLUMN course_credits;
```

```
INSERT INTO courses VALUES (1, 'Java', 4, 6);
UPDATE courses SET course_duration=8 WHERE course_id=1;
DELETE FROM courses WHERE course_id=1;
```

```
SELECT * FROM courses;
SELECT * FROM courses ORDER BY course_duration DESC;
SELECT * FROM courses LIMIT 2;
```

```
GRANT SELECT ON courses TO user1;
REVOKE INSERT ON courses FROM user1;
```

```

INSERT INTO courses VALUES (2,'Python',3,5);
COMMIT;

SAVEPOINT sp1;
UPDATE courses SET course_duration=10 WHERE course_id=2;
ROLLBACK TO sp1;

SELECT e.emp_name, d.dept_name
FROM employees e
INNER JOIN departments d
ON e.dept_id = d.dept_id;

SELECT d.dept_name, e.emp_name
FROM departments d
LEFT JOIN employees e
ON d.dept_id = e.dept_id;

SELECT dept_id, COUNT(*) FROM employees GROUP BY dept_id;
SELECT dept_id, AVG(salary) FROM employees GROUP BY dept_id;

CREATE PROCEDURE get_emp()
BEGIN
SELECT * FROM employees;
END;

CREATE VIEW emp_view AS
SELECT * FROM employees;

CREATE TRIGGER trg AFTER INSERT ON employees
FOR EACH ROW
INSERT INTO log_table VALUES ('Inserted');

BEGIN
DBMS_OUTPUT.PUT_LINE('Hello');
END;

BEGIN
IF 1=1 THEN
DBMS_OUTPUT.PUT_LINE('True');
END IF;
END;

DECLARE
CURSOR c IS SELECT * FROM employees;
BEGIN
FOR rec IN c LOOP
DBMS_OUTPUT.PUT_LINE(rec.emp_name);
END LOOP;
END;

SAVEPOINT sp1;
ROLLBACK TO sp1;

```